General Purpose Strategies for Handling Peak Overlaps in Atom Probe Data

Introducing the "Fractional Occupancy" Ranging Method

Olof Hellman[1] and David Reinhard[2]   [1]Tomographic LLC, Bothell, WA, United States.   [2]CAMECA Instruments Inc., Madison, WI, United States

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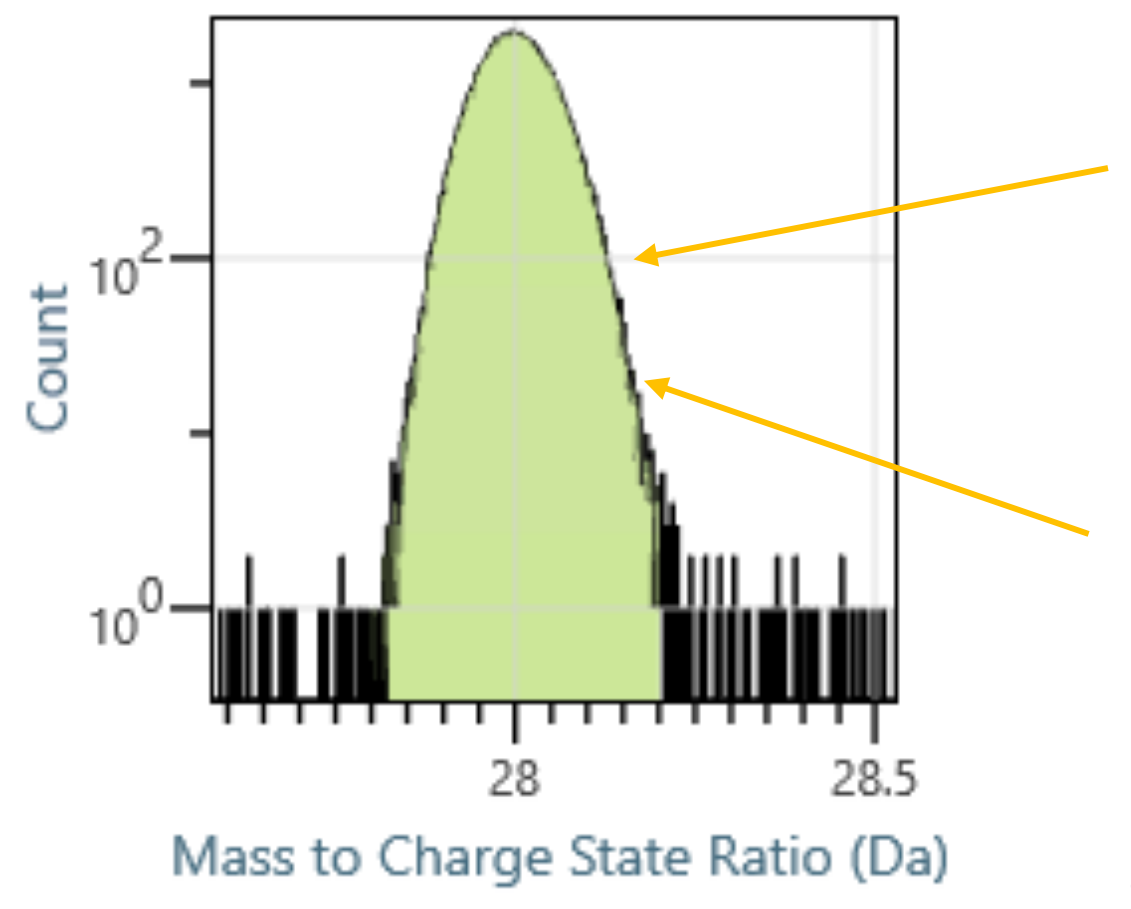
What is a "Range"?



Traditional Interpretation:
An ion in this peak corresponds to exactly 1 ion.

A .rrng file specifies exactly which ion in should be, i.e. one Si^+ ion, or one N_2^+, or one Fe^{++}, or whatever.

We refer to this as "integer occupancy"

Fractional Occupancy Interpretation:
An ion in this peak corresponds to some combination of fractional representation of one or more ions.

For example, it might correspond to 0.78 of a Si^+ ion, and 0.21 of a N_2^+ ion, and 0.01 percent background noise.

We refer to this as "fractional occupancy"

Our Motivation:
Implementing ranges with fractional occupancy should allow the effects of known peak overlaps and background noise to be meaningfully and more consistently incorporated into downstream analyses.

Implementation in IVAS

A) via RRNG file

Traditionally, the ion assignment for a particular range can be defined by a .RRNG file.

Here's how we propose extending the standard .RRNG file to specify occupancy:

```
[Ions]
Number=2
Ion1=Si
Ion2=N

[Ranges]
Number=5
Range1=27.8 28.2 Vol:0.02 Occupancy:0.78  Si:1 Color:CCCCCC
Range2=27.8 28.2 Vol:0.04 Occupancy:0.21  N:2  Color:33CCFF
Range3=28.8 29.2 Vol:0.02 Occupancy:0.99  Si:1 Color:CCCCCC
Range4=28.8 29.2 Vol:0.04 Occupancy:0.001 N:2  Color:33CCFF
Range5=29.8 30.2 Vol:0.02 Occupancy:0.99  Si:1 Color:CCCCCC
```

In the past, if a single range was used more than once in RRNG file, the exact behavior was undefined. IVAS has up until now worked on the "last one wins" principle.

B) via Peak Decomposition

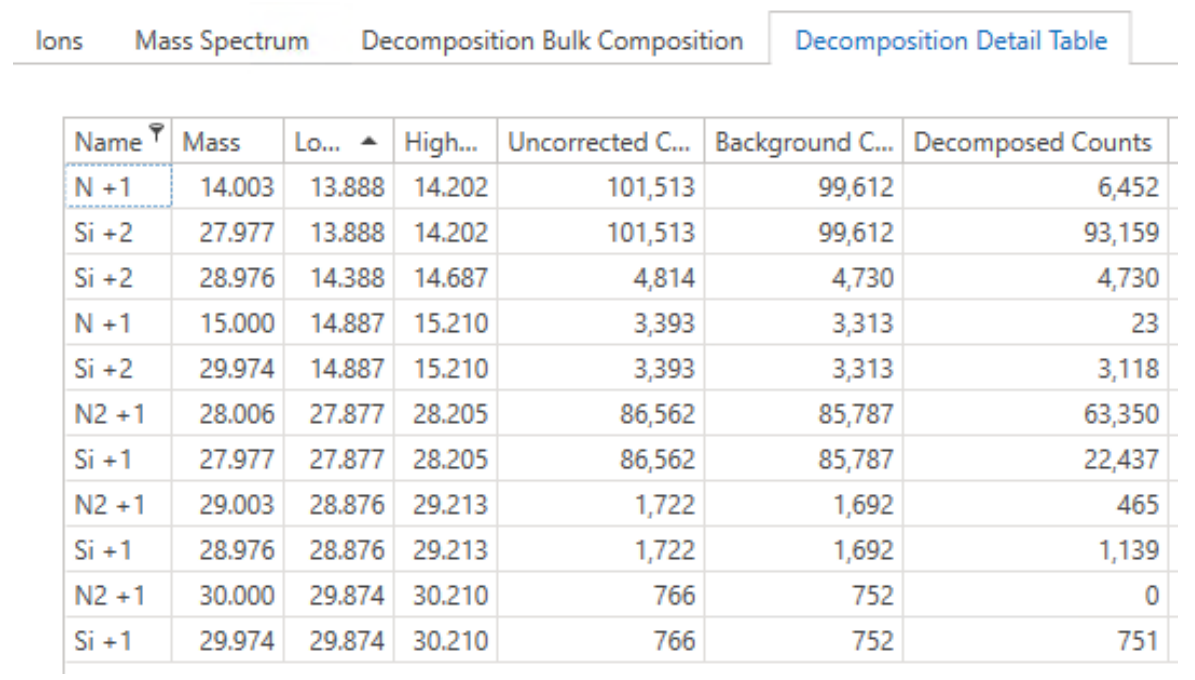
When a "Decomposition of Peaks" node is defined for a region of interest (ROI), the result of the decomposition is exactly the information required to define the fractional occupancies for ranges.

We explored implementing a strategy for defining occupancies by simply copying the Peak Decomposition fractions.

Any SubROI node or Detached ROI node can override its ranging by selecting an "Ion Build Strategy".

The "Peak Decomposition Verbatim" strategy applies the results of the Decomposition of Peaks to every range identified in the Decomposition.

The resulting ranges are displayed in the Mass Spectrum Analysis Pane.



Name	Mass	Lo...	High...	Uncorrected C...	Background C...	Decomposed Counts
N +1	14.003	13.888	14.202	101,513	99,612	6,452
Si +2	27.977	13.888	14.202	101,513	99,612	93,159
Si +2	28.976	14.388	14.687	4,814	4,730	4,730
N +1	15.000	14.887	15.210	3,393	3,313	23
Si +2	29.974	14.887	15.210	3,393	3,313	3,118
N2 +1	28.006	27.877	28.205	86,562	85,787	63,350
Si +1	27.977	27.877	28.205	86,562	85,787	22,437
N2 +1	29.003	28.876	29.213	1,722	1,692	465
Si +1	28.976	28.876	29.213	1,722	1,692	1,139
N2 +1	30.000	29.874	30.210	766	752	0
Si +1	29.974	29.874	30.210	766	752	751

Here, Decomposition of Peaks has identified:

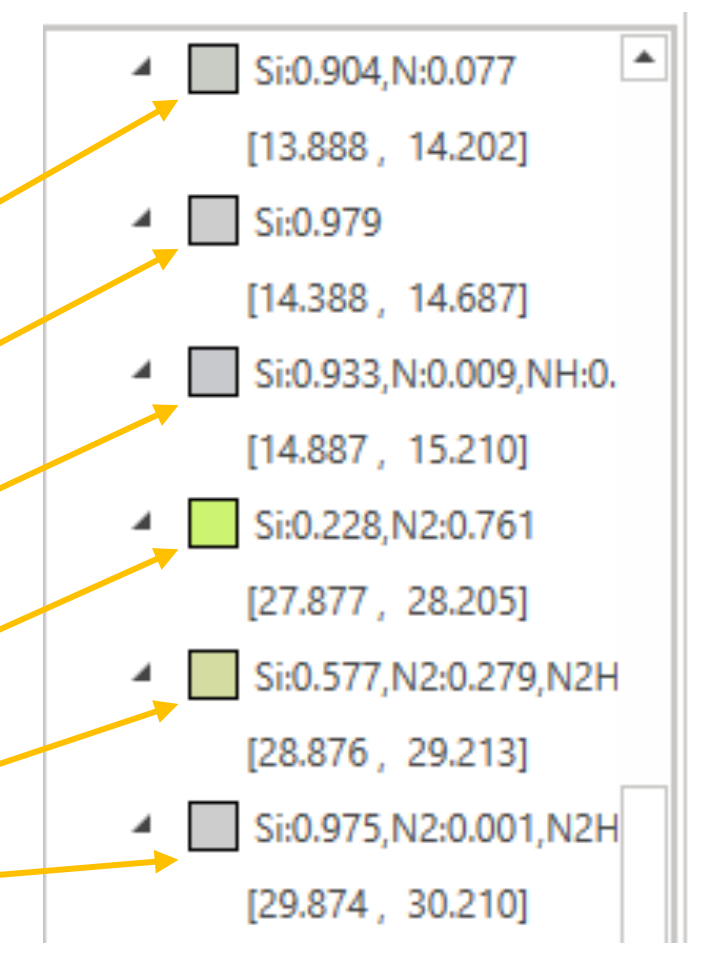
The peak at 14 Da is .904 Si and 0.077 N

The peak at 14.5 Da is 0.979 Si and 0.021 noise.

The peak at 15 Da is composed of 3 ions, Si, N and NH

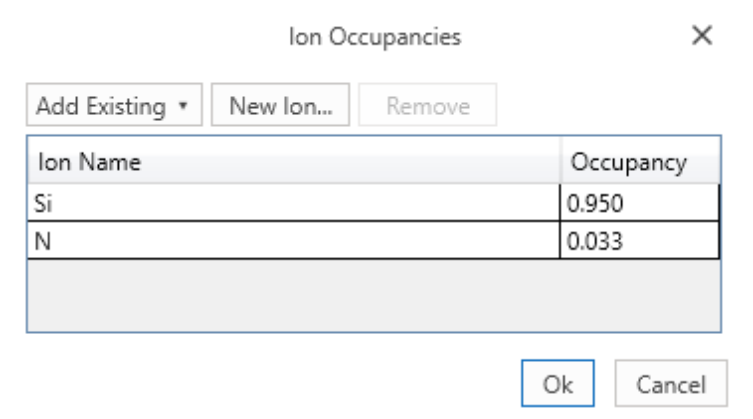
The peak at 28 Da is .228 Si and .761 N_2

The peaks at 29 and 30 Da are both composed of 3 ions, Si, N_2 and N_2H


Si:0.904,N:0.077
[13.888 , 14.202]
Si:0.979
[14.388 , 14.687]
Si:0.933,N:0.009,NH:0.
[14.887 , 15.210]
Si:0.228,N2:0.761
[27.877 , 28.205]
Si:0.577,N2:0.279,N2H
[28.876 , 29.213]
Si:0.975,N2:0.001,N2H
[29.874 , 30.210]

C) via Manual Editing

There's also a simple editor UI for adding fractional occupancies to a range



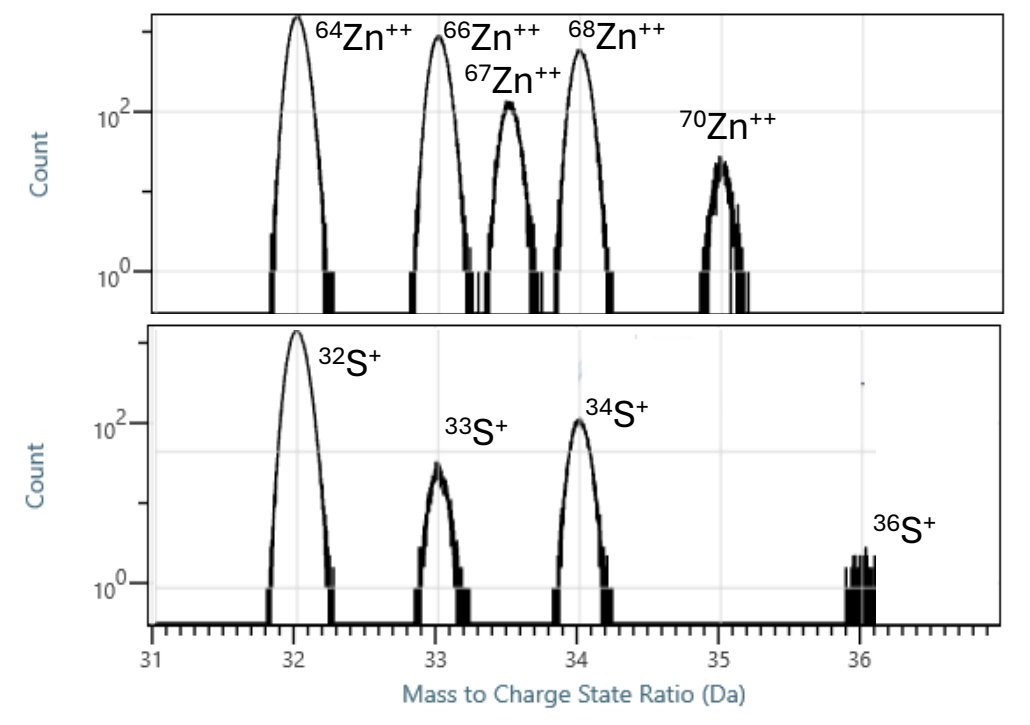
Ion Name	Occupancy
Si	0.950
N	0.033

Simulated samples

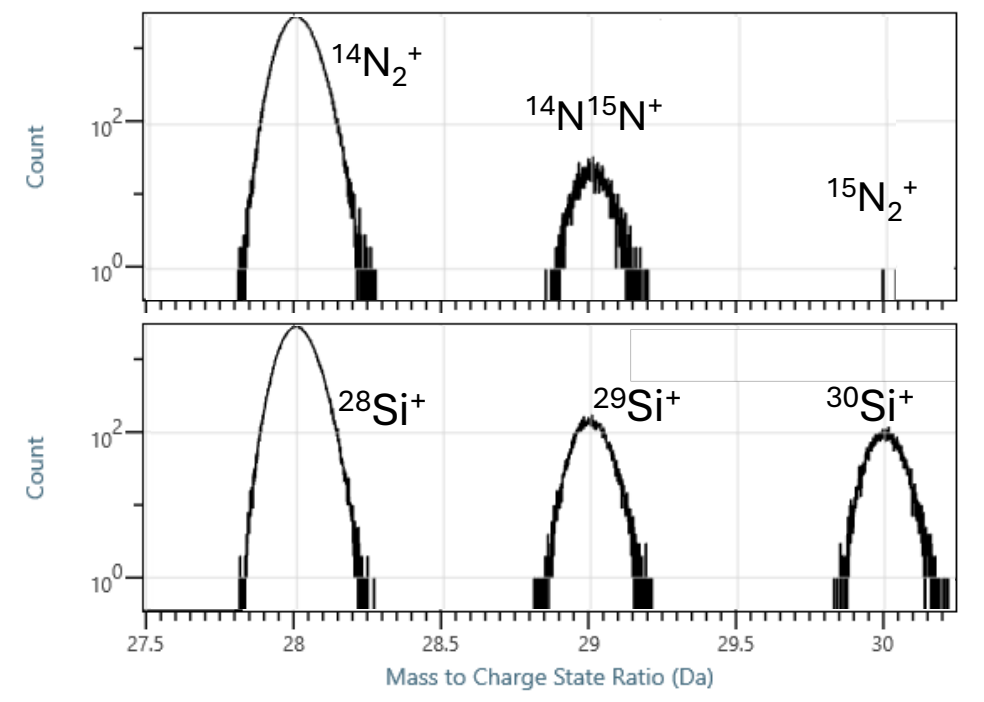
To demonstate the technique and test its limits, two series of datasets were made for systems that exhibit significant peak overlaps.

For both S / Zn and N / Si, majority components of both species overlap.

Zn and S Peak Overlap at 32 – 36 Da:



N and Si Peak Overlap at 28 – 30 Da:



ZnFeS spheres in FeS: 1 – 4 nm radius

A simulated sample of 10 ZnFeS spheres arranged in a BCC pattern in a matrix of FeS was made. The ZnFeS spheres have a 4-1 ratio of Zn to Fe. The ion populations used were:

FeS matrix:

Ion	%
Fe^+	25%
Fe^{++}	25%
S^+	40%
S^{++}	10%

ZnFeS precipitates:

Ion	%
Fe^+	5%
Fe^{++}	5%
Zn^+	20%
Zn^{++}	20%
S^+	40%
S^{++}	10%

Additionally, a 2% noise background was applied to the entire spectrum,decaying quadratically with increasing Da.

The resulting data should correspond exactly to FeS in the matrix, and to $Zn_{0.8}Fe_{0.2}S$ in the spheres.

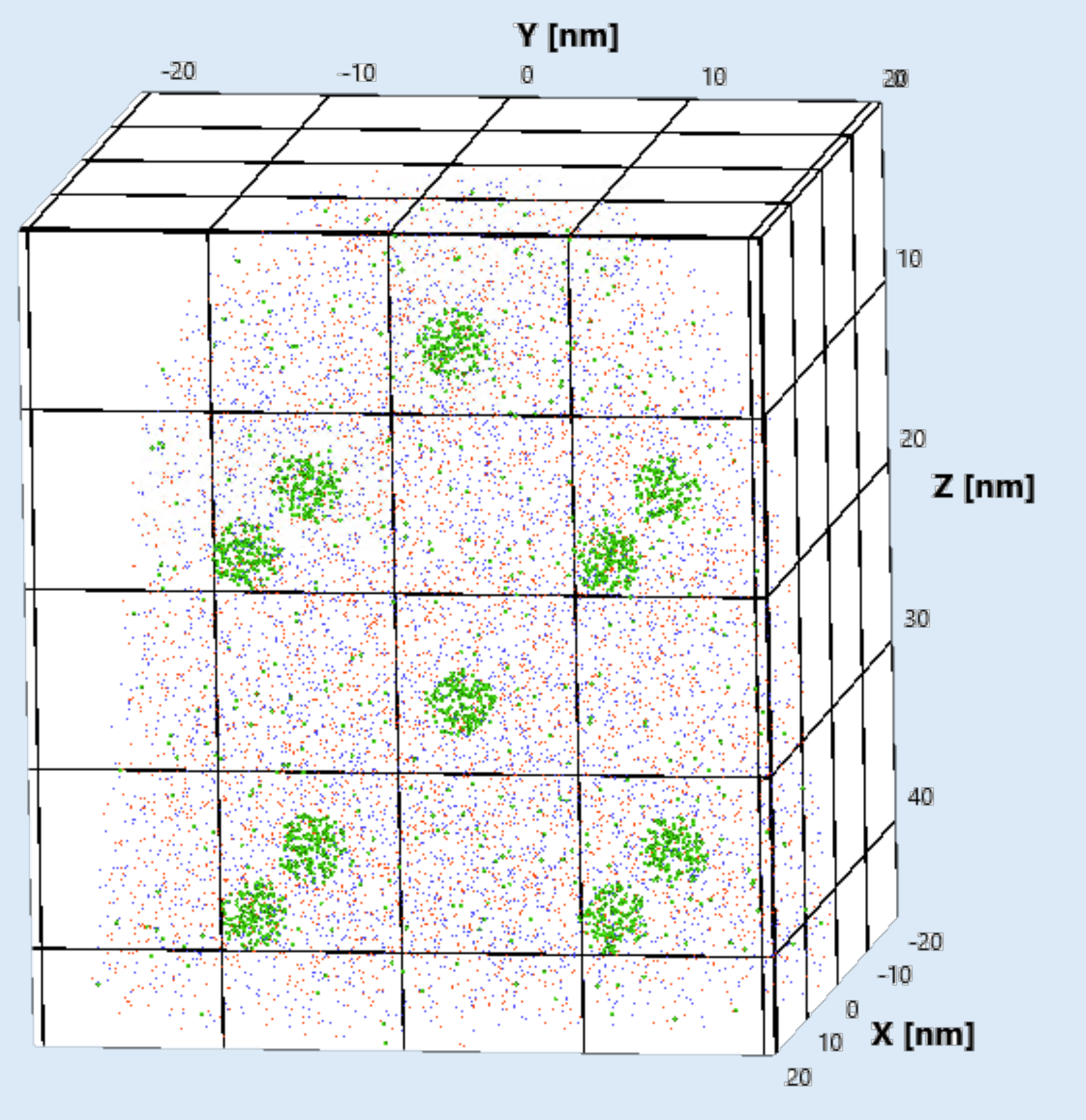
A range of simulated datasets was prepared ranging from 1 nm to 4 nm diameter spheres.

Data Analysis

After importing to IVAS, a .RRNG file was loaded corresponding to each component. No overlapping components were defined in the RRNG file. The only ranges assigned to Zn were those without S overlaps.

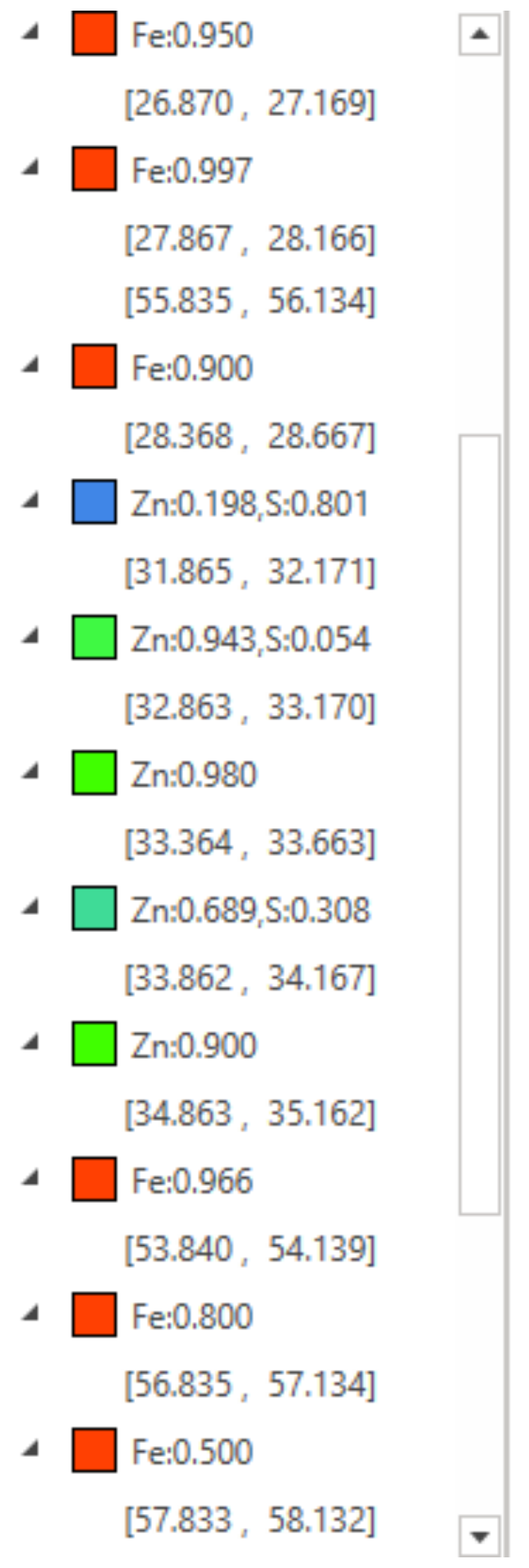
An isosurface was generated from the Zn ion counts, which was used to identify an ROI associated with the spheres. The isovalue was chosen at maximum slope of the proxigram.

Decomposition of Peaks was then used to supply fractional occupancies for each ROI.



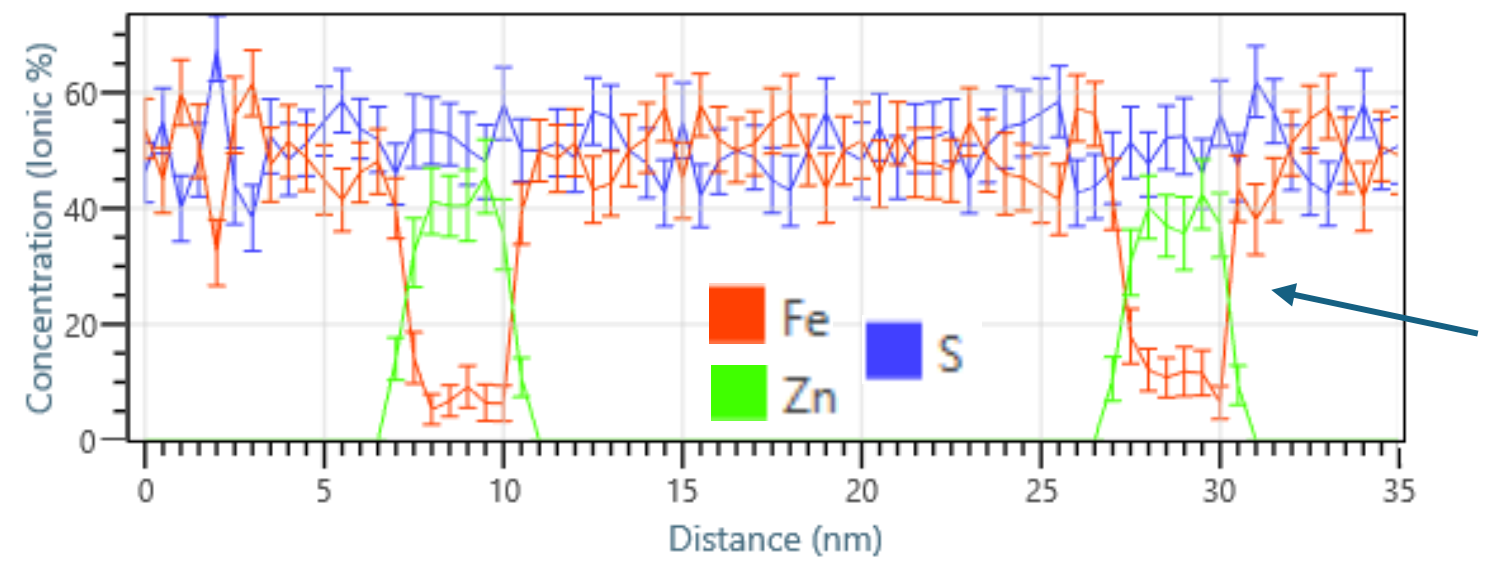
Ion map shown before any fractional occupancy ranges are defined.
Zn ions shown in green are from Zn peaks that have no overlaps with S.
1% of the Fe and S ions are displayed
Spherical clusters are arranged in a BCC pattern

After defining an isosurface and creating an ROI for the spheres, Decomposition of Peaks produces new ranges. Shown here ranges for the 4nm diameter sample:


Fe:0.950
[26.870 , 27.169]
Fe:0.997
[27.867 , 28.166]
[55.835 , 56.134]
Fe:0.900
[28.368 , 28.667]
Zn:0.198,S:0.801
[31.865 , 32.171]
Zn:0.943,S:0.054
[32.863 , 33.170]
Zn:0.980
[33.364 , 33.663]
Zn:0.689,S:0.308
[33.862 , 34.167]
Zn:0.900
[34.863 , 35.162]
Fe:0.966
[53.840 , 54.139]
Fe:0.800
[56.835 , 57.134]
Fe:0.500
[57.833 , 58.132]

After applying new ranges, one dimensional profiles through the two center spheres: (cylinder diameter ~ 0.75 identified sphere diameter)

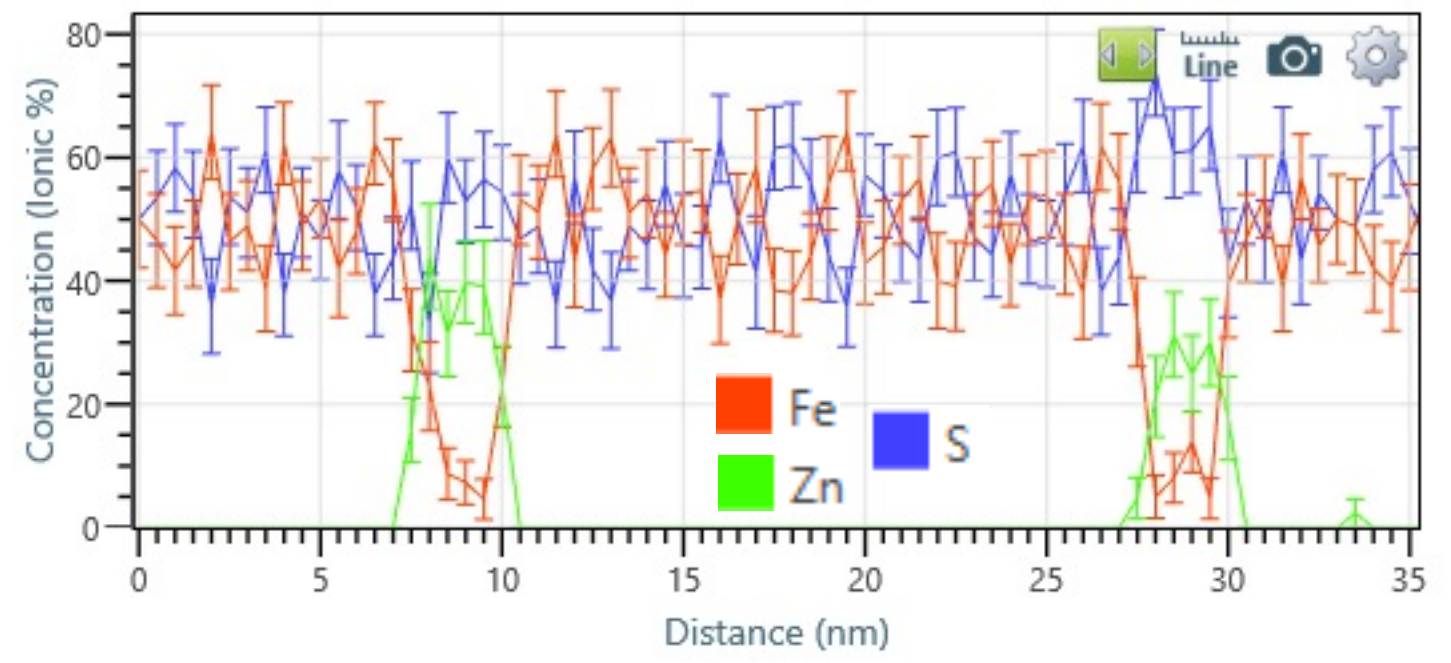
4 nm diameter:



In the case of the 4 nm diameter spheres, the calculated compositions of the spheres are as expected:

0.5 S, 0.4 Zn and 0.1 Fe

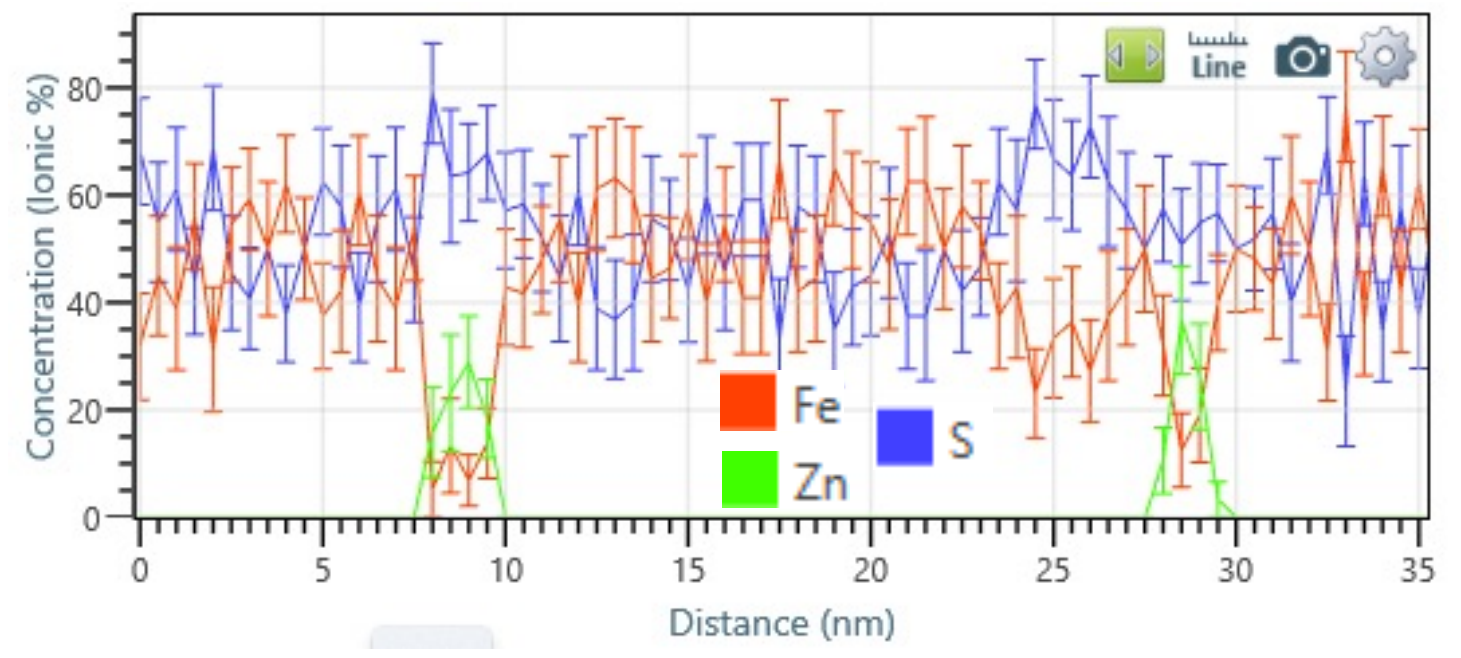
3 nm diameter:



For the 3 nm diameter spheres, the calculated compositions of the spheres begins to degrade.

There's an (incorrect) increase in identified S inside the spheres, and a corresponding decrease (also incorrect) in the identified Zn.

2 nm diameter:



For the 2 nm diameter spheres, the statistics of the composition calculation begin to get challenging.

The statistical error applies both to the composition calculation in the 1D profile, but also in the Peak Decomposition algorithm

There is still error in the identified S inside the spheres, and a corresponding decrease in the identified Zn. There is also an anomaly of unknown origin adjacent to one sphere at about distance = 25 nm

SiN / Si with varying layer thickness

Simulated samples of SiN and Si layers were made with ion populations similar to a real sample. The ion fractions used were,

Si layers:

Ion	%
Si^+	25.8%
Si^{++}	73.6%
H^+	0.4%
H_2^+	0.1%

SiN layers:

Ion	%
Si^+	10.1%
Si^{++}	53.2%
N_2^+	30.8%
SiN^+	1.1%
Si_2N^{++}	1.4%
N^+	2.8%
N_2H^+	0.14%
NH^+	0.03%
SiN^+	1.1%
H^+	0.28%
H_2^+	0.05%

Additionally, a 2% noise background was applied to the entire spectrum, decaying quadratically with increasing Da.

The resulting data should correspond exactly to 100% Si in the Si layers, and to 50% Si / 50%N in the SiN layers if only the Si and N atoms are considered.

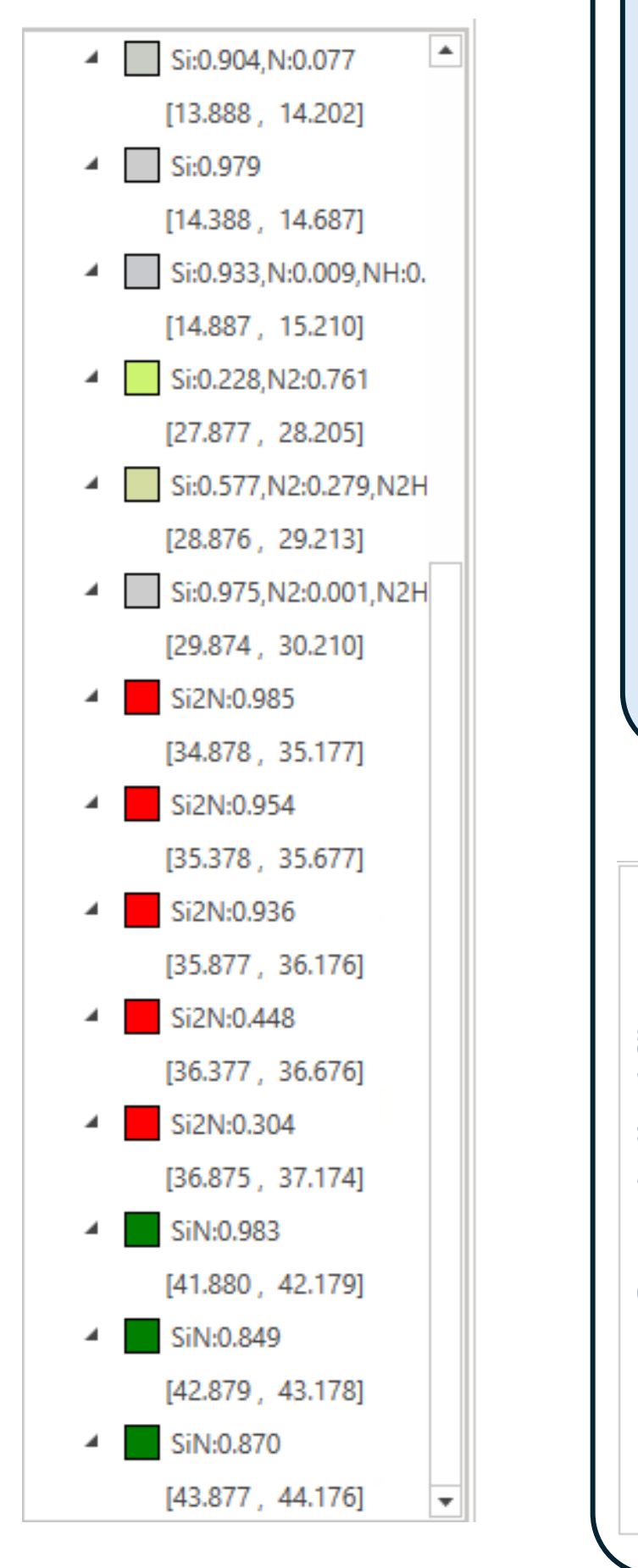
A range of simulated datasets was prepared ranging from 1nm thickness in each layer (2nm period) to 8 nm (16 nm period).

Data Analysis

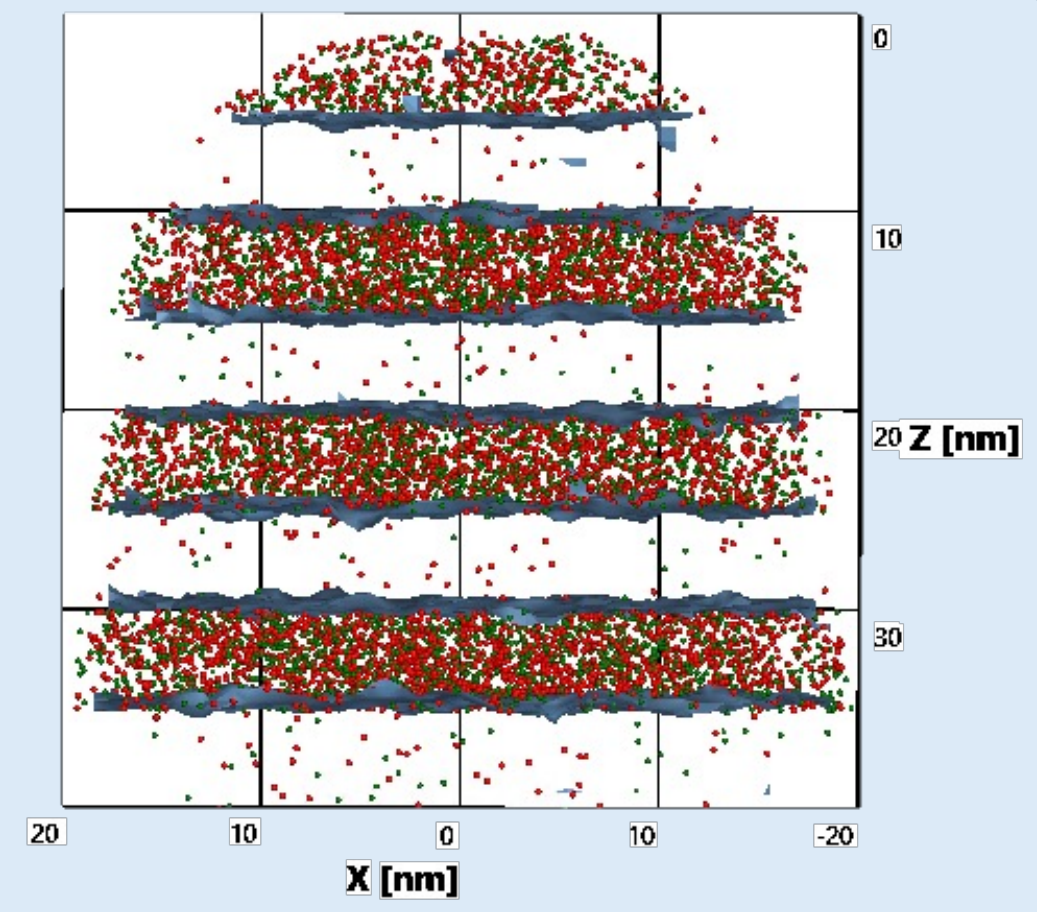
After importing to IVAS, a .RRNG file was loaded corresponding to each component. Overlapping components were assigned to be equally divided between each component. (i.e.,the 28 Da peak was assigned to be 50% N_2^+ and 50% Si^+)

An isosurface was generated from the SiN^+, Si_2N^{++} and N_2^+ counts, which was used to partition the sample into two ROIs. Decomposition of Peaks was used to supply fractional occupancies for each ROI.

After ROI definition and Decomposition of Peaks, fractional occupancies are automatically assigned to ranges using the Peak Decomposition results, resulting ranges like these:


Si:0.904,N:0.077
[13.888 , 14.202]
Si:0.979
[14.388 , 14.687]
Si:0.933,N:0.009,NH:0.
[14.887 , 15.210]
Si:0.228,N2:0.761
[27.877 , 28.205]
Si:0.577,N2:0.279,N2H
[28.876 , 29.213]
Si:0.975,N2:0.001,N2H
[29.874 , 30.210]
Si2N:0.985
[34.878 , 35.177]
Si2N:0.954
[35.378 , 35.677]
Si2N:0.936
[35.877 , 36.176]
Si2N:0.448
[36.377 , 36.676]
Si2N:0.304
[36.875 , 37.174]
SiN:0.983
[41.880 , 42.179]
SiN:0.849
[42.879 , 43.178]
SiN:0.870
[43.877 , 44.176]

5 nm



Isosurface defined for 5 nm Si/SiN layers (pre-decomposition)
Exact isovalue chosen to minimize noise between interfaces.
Interface definition is good. Only SiN^+ and SiN_2^{++} ions shown.

Before identification of the fractional occupancies of each range, there is enough information to partition the sample into two ROIs.

Once the ROIs are defined, Decomposition of Peaks operates based on just the ions in the ROI.

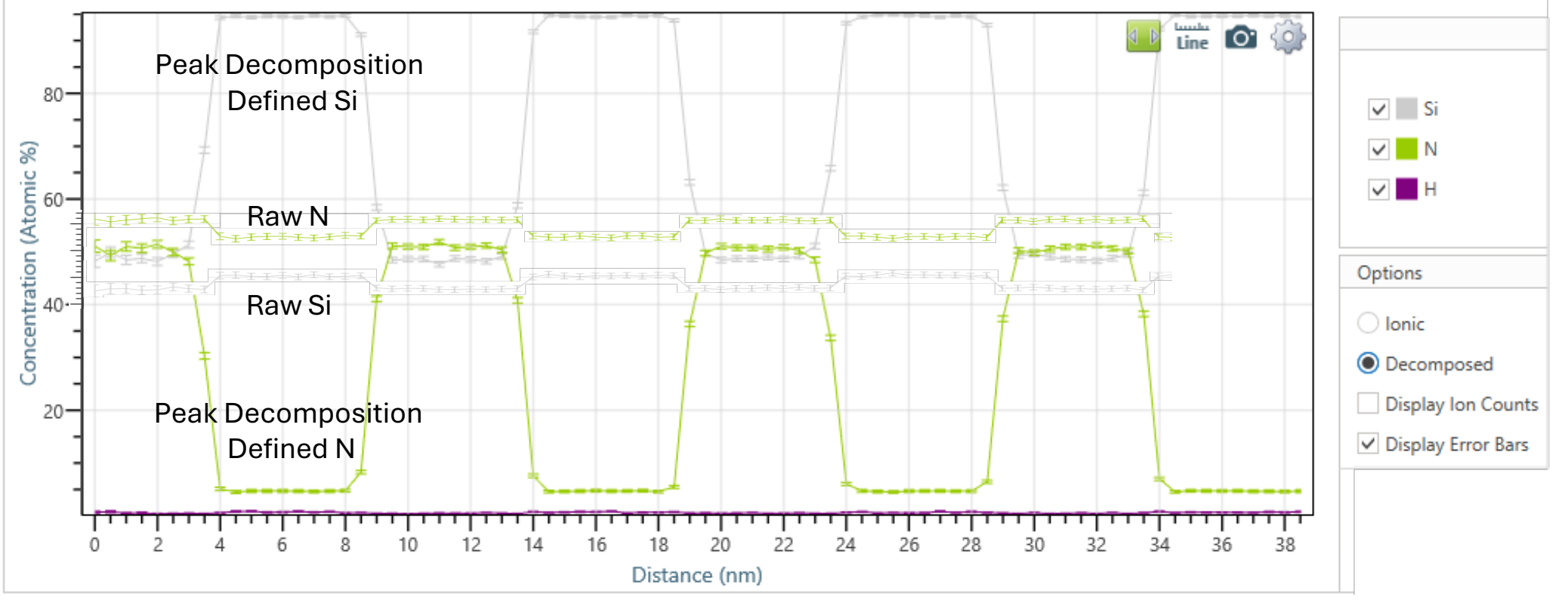
The resulting peak decompositions are used to define the fractional occupancies of each range identified in the decomposition.

After assigning fractional occupancies, the 1D profile for the 5nm layer sample shows good agreement with the expected compositions. A cylinder ROI was used for the profile to avoid edge effects from most of the sample.

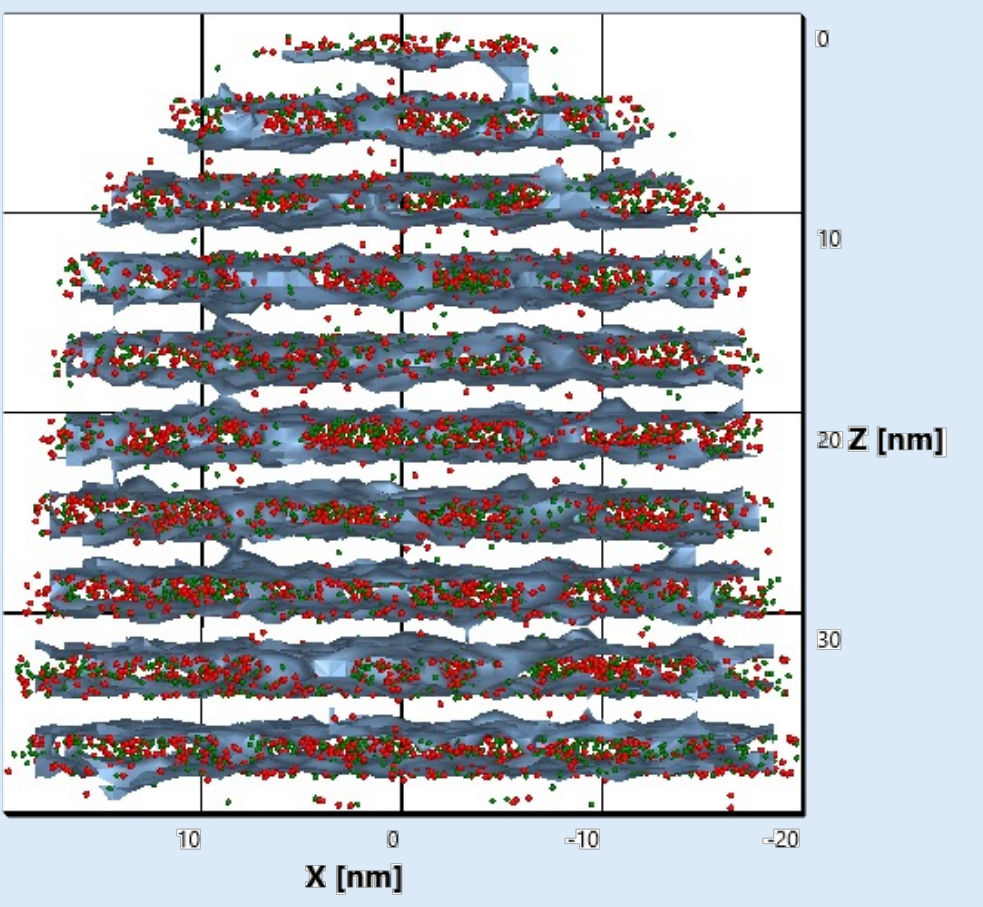
The errors (a small level of N in what should be pure Si regions) and not-quite 50-50 split in the SiN regions are likely due to imprecise definition of the isosurface separating the two ROIs.

The profile below compares the compositions before and after adjusting for the peak decomposition

One dimensional profile through sample, with overlay of raw (overlap peak 50-50) and peak-decomposition-adjusted ranges



2 nm



Isosurface defined for 2 nm layers. Exact isovalue chosen to minimize noise between interfaces. Interface definition is becoming poorer. Only SiN^+ and SiN_2^{++} ions shown

For the 2nm layers sample, the isosurface definition has lost some fidelity, and the delineation between the two ROIs is more difficult.

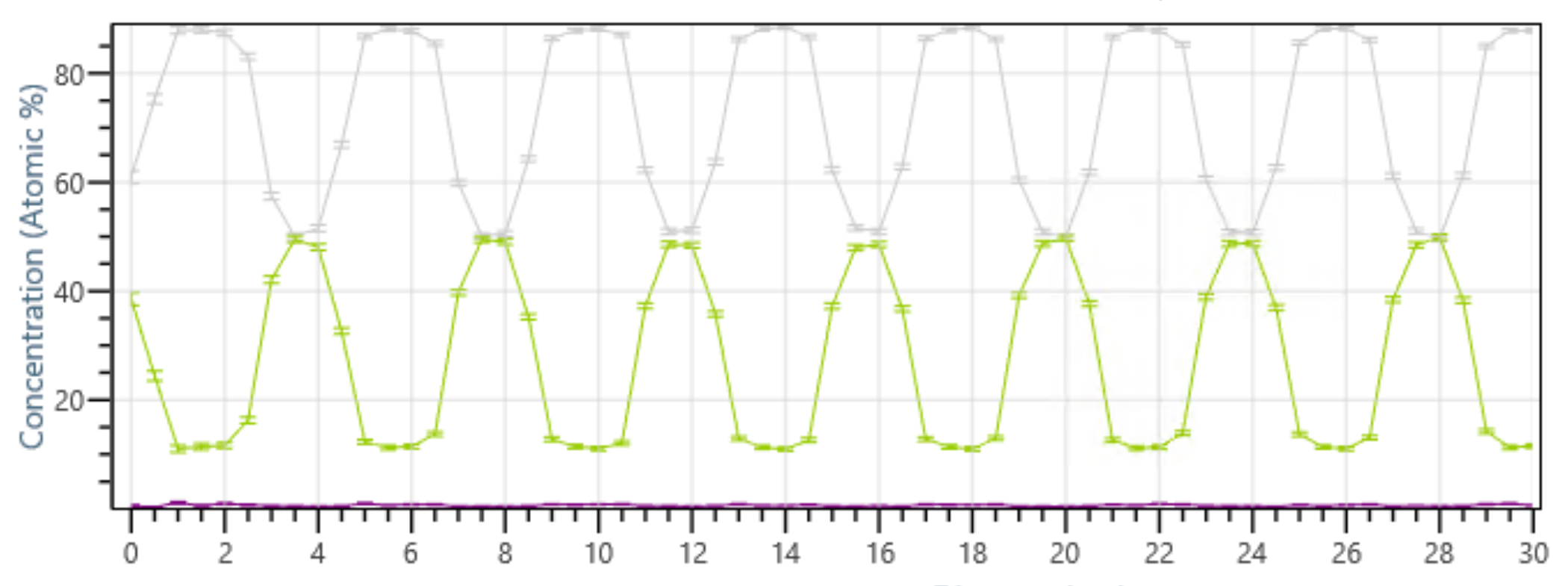
After assigning the fractional occupancies, the one dimensional profile for the 2nm layer sample shows worse agreement with the expected compositions.

While the SiN layer is still identified as being approximately 50% Si and 50% N, the fidelity is slightly worse than for the thicker-layer samples. The errors in identifying the pure Si regions have increased.

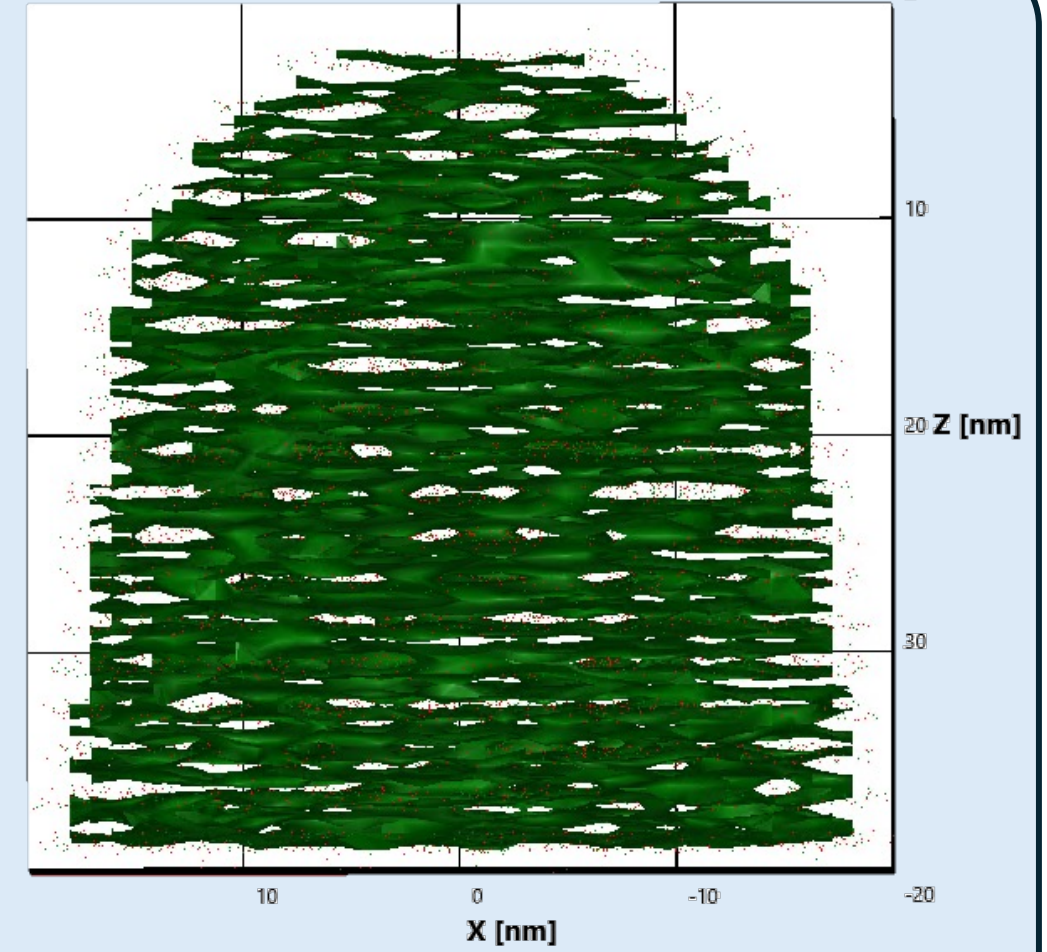
Conclusion:

The most important factor for success in using Decomposition of Peaks for fractional occupancy assignments is a clean delineation of the ROIs

One dimensional profile through sample, showing only composition calculated from peak-decomposition-adjusted ranges



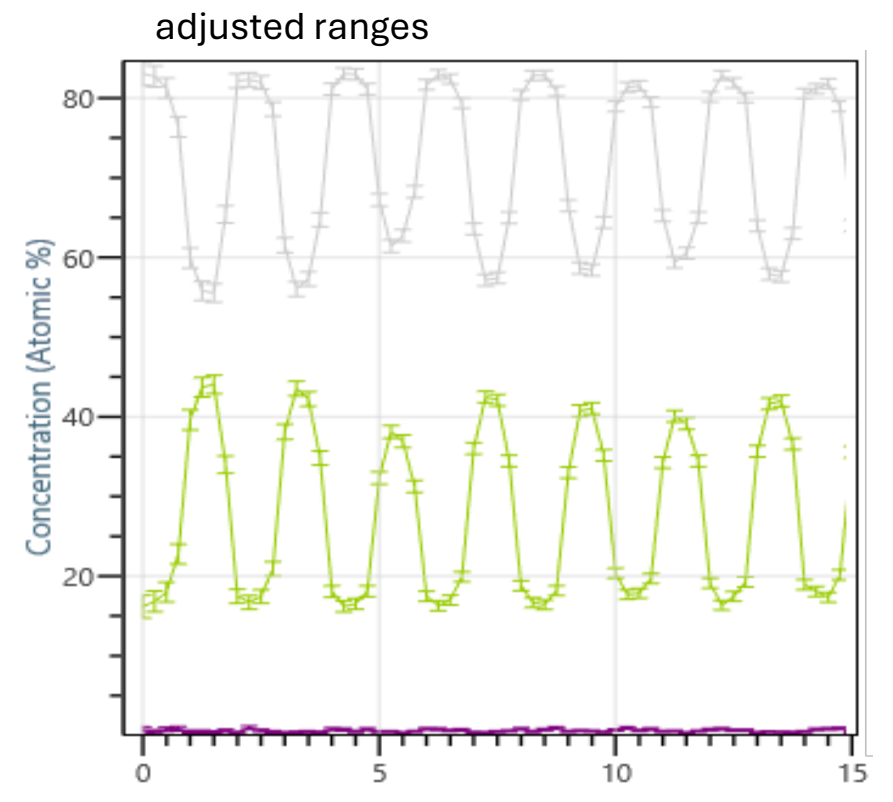
1 nm



Isosurface defined for 1 nm layers. Exact isovalue chosen to minimize noise between interfaces. Interface definition is very poor. Only SiN^+ and SiN_2^{++} ions shown

One dimensional profile through sample, adjusted ranges

For 1 nm layers, the ROI delineation is poor, and the layer compositions are no longer close at all to expected values.



Software Considerations

"Integer Occupancy" is an assumption in many of the algorithms used for processing ranged data. However, conversion to fractional occupancy is straightforward in many cases.

Grid concentrations, ratio fractions, composition profiles, composition maps, and proxigrams can all be easily derived from fractional occupancy data. The main change is accumulating floating point values instead of integer values.

However, some analysis techniques are not as easy to convert for example, Cluster Identification algorithms use neighbor calculations to connect proximate ions of a certain type. It is not obvious how to extend such algorithms to fractional occupancy.

Another situation where integer occupancy is convenient is in rendering datasets. We have tried different ways to render fractional occupancy ion maps, but the current methods used for integer occupancy and seem to be the most straightforward.

Fractional occupancy easily supports backwards compatibility. That is, where previously a range was defined associating a range to an integer ion now asscoiates that range to a "fractional" occupancy of 1.0. All the successive calculation should be unchanged.

User Input Needed (i.e. your input needed)

We are very interested in how fractional occupancy might impact other more esoteric analysis techniques. And we are very sensitive to the needs of users of software when attempting to implement such a feature.

Does "fractional occupancy" as a software feature make sense? Are there better terms for describing how it works?

How should fractional occupancy range be displayed and edited in the user interface? Will users understand where the fractional occupancy values came from when using Decomposition of Peaks to set the values? Will users understand the downstream implications of fractional occupancy ranging?

How might the statistical error in calculating the fractional occupancies be made visible to users? Is it important to propagate the error through other calculations?

In short, what other software support will users need to take full advantage of this feature?

Deriving an integer-occupancy dataset

When needed, a fractional occupancy dataset can be converted to an integer occupancy dataset if needed, but with some dataloss.   The algorithm we use is an accumulative model, which we use for rendering the ion maps:

The goal of the algorithm is to assign each datapoint an integer ion. Each candidate Ion with fractional occupancy is assigned a "remainder", initialized at 0.5.

The algorithm is iterative: For each ion coordinate in the range:

A) each remainder is incremented by the amount of its Ion's occupancy.

B) if the Candidate Ion with the highest remainder has a remainder of greater than or equal to 1.0, that Ion is chosen for that datapoint, and its remainder is reduced by 1.0. It is possible that no Ion has remainder of higher than 1.0, in which case that datapoint is considered 'background'.

Here's a table that shows how the process works for 15 ion coordinates in a range with IonA at 0.7 occupancy, IonB at 0.25 occupancy (background is 0.05),

Rem A	RemB	Ion
0.5	0.5	A
0.2	0.75	B
0.9	0	A
0.6	0.25	A
0.3	0.5	A
0	0.75	B
0.7	0	A
0.4	0.25	A
0.1	0.5	
0.8	0.75	A
0.5	1	B
1.2	0.25	A
0.9	0.5	A
0.6	0.75	A
0.3	1	B
1	0.25	A
0.7	0.5	A
0.4	0.75	A
0.1	1	B
0.8	0.25	A
0.5	0.5	A
0.2	0.75	B
0.9	0	A
0.6	0.25	A
0.3	0.5	
1	0.75	A

The first ion is determined to be A. RemainderA is incremented to 1.2, and RemainderB is incremented to 0.75. 1.2 is largest and greater than 1, so the ion is A and RemainderA is decremented to 0.2. The process continues iteratively for each datapoint, with the previous remainders used as input for the next cycle.

In this example, the 9th and 25th data points are not drawn – they represent background ions. Over a larger sample, exactly 5% of the coordinates will be background, 75% of the datapoints will be IonA, and 20% of the datapoints will be IonB.

This strategy is deterministic, in that it produces the same output every time given the same input.

There is also room for manipulation of the initial remainder values to generate a different deterministic output from the same input. That is, if different values are used for the initial remainders, different ion assignments will be made. This could be useful for multiple runs of a cluster search algorithm using fractional occupancy input data.

Previous work

Much work has been done on spatially resolved peak decomposition, and indeed, various strategies for applying peak deconvolution to improve experimental results have been successful. The references listed here are only a small sampling.

What is new in this work is implementing a fractional occupancy ranging treatment as a **general purpose** feature of software (see the title of the poster!).

That is, there may be many ways to define regions of interest, or to apply different fractional occupancies, based on spatial techniques. Many, if not all, of these techniques, could use a fractional occupancy approach to leverage those techniques.

We hope that our proof-of-concept in integrating fractional occupancy into the IVAS platform will be enabling for everyone trying to overcome peak overlap challenges in APT data

References:

Kuduz, M., Schmitz, G., and Kirchheim, R. (2004). Investigation of oxide tunnel barriers by atom probe tomography (tap). Ultramicroscopy, 101(2–4):197 – 205.

London AJ, Lozano-Perez S, Moody MP, Amirthapandian S, Panigrahi BK, Sundar CS, Grovenor CR. Quantification of oxide particle composition in model oxide dispersion strengthened steel alloys. Ultramicroscopy. 2015 Dec;159 Pt 2:360-7.

London AJ, Haley D, Moody MP. Single-Ion Deconvolution of Mass Peak Overlaps for Atom Probe Microscopy. Microsc Microanal. 2017 Apr;23(2):300-306.

Engberg DLJ, Johnson LJS, Jensen J, Thuvander M, Hultman L. Resolving mass spectral overlaps in atom probe tomography by isotopic substitutions - case of TiSi15N. Ultramicroscopy. 2018 Jan;184(Pt A):51-60.

London AJ. Quantifying Uncertainty from Mass-Peak Overlaps in Atom Probe Microscopy. Microsc Microanal. 2019 Apr;25(2):378-388.

Conclusions

The fractional occupancy ranging strategy is both useful for handling peak overlaps in atom probe data and can be applied to any system that has peak overlaps. It may also be useful simply as a strategy for handling background subtraction.

When using the Decomposition of Peaks method for automatic setting of occupancy fractions, it is important to be able to define regions of interest appropriately and consistently. We show that quality of the data analysis suffers when the regions of interest cannot be delineated well.

Our proof-of-concept implementation of fractional occupancy in the IVAS demonstrates feasibility as a general purpose technique, able to be integrated with other APT data analysis techniques.

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